

1 Implementation Guide: BDE

This section describes how HYDRA-BDE interacts with DMC. This section is based on the *DMC Implementation Guide*.

Further applicable documents

The following documents deal with DMC basics or add to the contents of this document:

- DMC Implementation Guide
- DMC Implementation Guide MDE
- DMC-SDK (delivered with DMC)
 - o DMC-API documentation
 - o Sample ImplementationGuide

The *SDK Sample ImplementationGuide* provides examples on how BDE integrates in the reference process.

Data posted onto the operation

The DMW (Dynamic MES Weaver) controls the production process of items/parts on the production line. Every physical item is represented by a digital workpiece. The operation is the basis for manufacturing an item.

Once production is completed and depending on the production result, the DMW posts a yield or scrap quantity for the operation. Configure the following components in the *factory model* to enable this function.

Configure the component *ProductionManager* for all stations, i.e. add an instance as child to the relevant station.

```
<ComponentConfiguration
  Name="ProductionManager"
  Class="mpdv.MachineDataCollector.Dmc.Production.ProductionManager"
  Children="HydraMDEFileWriter" />
```

Assign a *HydraMDEFileWriter* component as child to the *ProductionManager* component in order to transfer postings to HYDRA. Refer to the *DMC-API* documentation and the *Implementation Guide MDE* for further information on these components.

Instead of using the default *ProductionManager* component, you can develop and use an alternative component to integrate BDE.

The digital workpiece stores the production period and place of production (station). The digital workpiece includes detailed information on each process and work step (BeginOfManufacturing and EndOfManufacturing).

Collection of personnel times

If staff is logged on to a workstation, the digital workpiece records for every work step:

- the staff involved in production and
- the period of time.

The component *PersonnelManager* collects this information. Configure this component for every station:

```
<ComponentConfiguration Name="PersonnelManager"
  Class="mpdv.MachineDataCollector.Dmc.Production.PersonnelManager">
  <Parameter Name="PersonRepositoryComponent" Value="PersonsDataRepository" />
</ComponentConfiguration>
```

Set an instance of a component as value for the parameter *PersonRepositoryComponent*. This component must provide objects including the field *PersonId*. This can be, for example:

```
<ComponentConfiguration Name="PersonsDataRepository"
  Class="mpdv.MachineDataCollector.Dmc.Model.SimpleDataRepository">
  <Parameter Name="ReadConnector" Value="JtpConnector" />
  <Parameter Name="ListServiceName" Value="BOPerson.list" />
</ComponentConfiguration>
```

The digital workpiece status stores personnel times as *EventInfo*. You can send login and logoff details to the capability *PersonnelManager* of the component *PersonnelManager*. Use the inputs *LogOnPerson* and *LogOffPerson* that transfer the *PersonId*. Refer to the *DMC-API* documentation for further information.